

Udayan Atreya

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EDUCATION

Master of Science, Computer Science

May 2026

San Jose State University, San Jose, US

Coursework: Applied Deep Learning, Retrieval Augmented Generation (RAG), Biometric Security with AI.

Bachelor of Technology, Computer Science & Engineering

May 2020

Manipal University, Jaipur, India

Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming.

TECHNICAL SKILLS

Web & Backend: FastAPI, REST APIs, .NET, AWS API Gateway, Lambda, Microservices, Git

Cloud & Database: AWS, S3, CI/CD, MongoDB, DynamoDB, ChromaDB, Pinecone, Azure DevOps

AI & ML: LangChain, RAG, HuggingFace, TensorFlow, PyTorch, RAGAS, GPT, Gemini, Claude

Languages: Python, Java, C#, SQL

PROJECT EXPERIENCE

Agentic BMC Observability Platform (*FastAPI, LangChain, Gemini 1.5, MongoDB, AWS S3*)

[\[GitHub Link\]](#)

Jun 2025 – Jul 2025

- Built a **FastAPI web platform** with real-time Grafana dashboards and well-documented REST endpoints, achieving **95% query classification accuracy** and **reducing manual triage time by 30%**.
- Designed and integrated **RESTful backend APIs** connecting a hybrid MongoDB/S3 data layer to platform consumers, cutting **data retrieval latency by 75%** (from 2 minutes to 30 seconds).
- Implemented role-based access controls and human-in-the-loop validations across API endpoints, **preventing 100% of unauthorized state changes** in production.

Evaluator-Optimizer RAG (*LangChain, GPT-4o, ChromaDB, RAGAS, Whisper*)

[\[GitHub Link\]](#)

Aug 2025 – Mar 2026

- Architected an agentic RAG framework with **clean, modular, well-documented code**, reducing hallucination by **25%** and improving answer accuracy by **18%** over baselines.
- Engineered an evaluation suite using RAGAS and LLM-as-a-judge on 500+ STEM QA pairs, quantifying Faithfulness, Answer Relevancy, Context Precision, and Semantic Similarity.
- Optimized latency-accuracy trade-off via Dynamic Chunking and MMR search, reducing **context redundancy by 40%** while maintaining over **90% semantic coverage**.

WORK EXPERIENCE

Software Engineer, Accenture, Bengaluru, India

Dec 2021 – Jul 2024

Associate Software Engineer, Accenture, Bengaluru, India

Nov 2020 – Nov 2021

- Architected a **user-facing notification service** integrating **AWS API Gateway and Lambda** with client applications via REST, resulting in a **20% increase in DAU** post-launch.
- Delivered a **web-based ChatOps platform** connecting a natural language frontend to backend serverless automation workflows through **REST API integration**, **reducing overhead by 40%**.
- Optimized **DynamoDB** schema and CRUD REST microservices for **high-performance data exchange**, **reducing API response latency by 37%** and ensuring scalable, resilient backends.
- Built a code vulnerability remediation solution using serverless microservices (Lambda, S3, EventBridge) and Azure DevOps CI/CD pipelines, safeguarding **400+ applications** with clean, maintainable code.
- Designed a server patching solution integrating ServiceNow alerts with serverless automation, **reducing MTTR by 80%**; collaborated with **cross-functional global teams** throughout delivery.